

When Financial Agents Hallucinate Alpha: Auditing Backtest Leakage, Compliance Paradoxes, and Social Megaphone Bias with Executable Guards

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Abstract

+0.0186).

Large language model (LLM) agents increasingly automate quantitative financial research, yet their reported backtest Sharpe ratios routinely collapse out-of-sample. We demonstrate that autonomous financial agents fail through two orthogonal mechanisms: **Type I Mechanical Shortcuts** (same-session post-close joins, pre-training window contamination, survivorship filtering, and unadjusted corporate actions) and **Type II Empirical & Microstructure Illusions** (trusting retail social echo-chambers where high-reach influencers carry negative alpha, and violating $T + 1$ settlement or board-lot constraints). To address both, we introduce **fin-skills**, an open-source library of **129 source-verified skills**, **36 executable verification guards** (`Bundle.check()`), and **53 MCP agent tools**, paired with an external **Counterfactual Audit Protocol** that grades agent submissions without using the library to grade itself. Across synthetic worlds with 12 planted defects (honest ceiling Sharpe 1.63 vs. same-day short-cut 71.5), our guards achieve **100% recall (36/36) with zero false alarms**, while physical session-by-session clock replay exposes leaky pipelines that score Sharpe 22.80 on held-out static years yet lose -1.17 bps/day (Sharpe -0.58) live. In a multi-model agent evaluation, **fin-skills** reduces strong-model (Claude Opus) Sharpe reporting error by **84%** ($0.19 \rightarrow 0.03$), but uncovers a striking **Compliance Paradox**: giving weak models (Claude Haiku) passive skill documentation alone yields the worst distortion (11.95 Sharpe gap) as agents *hallucinate citations to guards they never executed*. Finally, auditing **1.72M bilingual posts and 1,445 A-share KOL track records** reveals a $3.96\times$ follower asymmetry that drives unweighted NLP into negative correlation (Rank IC = -0.0318 , $p = 1.74 \times 10^{-20}$) until reversed by Bayesian credibility gating (Rank IC = $+0.0104$, $p = 0.001$) and LLM semantic distillation ($+0.0084 \rightarrow$

1 Introduction

Equipped with code interpreters and financial data feeds, LLM agents can formulate hypotheses, engineer alpha signals, and report backtest performance in minutes [5–7]. Yet in quantitative finance, high backtest Sharpe ratios almost never survive live capital deployment [1–3]. Through systematic auditing of LLM research trajectories, we show that agentic backtest inflation stems from two distinct failure classes:

1. **Type I: Mechanical Research Shortcuts.** Agents silently exploit structural artifacts in tabular workspaces: joining post-close news commentary to the same session’s open-to-close return, training on pre-computed LLM sentiment scores whose fitting window overlaps the evaluation period, filtering universes by future survival, or computing factors on unadjusted split prices.
2. **Type II: Empirical & Microstructure Illusions.** Even when timestamps are strictly causal, agents misinterpret noisy social streams and market rules. On financial social platforms, viral retail sentiment peaks at local price tops, while real execution is governed by $T+1$ settlement lockups, 100-share board lots, asymmetric sell-side stamp duties, and secondary-market ETF premium caps.

Inspired by behavioral testing in NLP [4] and modular verification systems [8], we present **fin-skills** paired with an external **Counterfactual & Microstructure Audit Protocol**. Rather than claiming to manufacture market alpha, our framework enforces verifiable research integrity. Our contributions are fourfold:

- **The fin-skills Verification Library (§2):** 129 source-verified skills, 36 executable guards

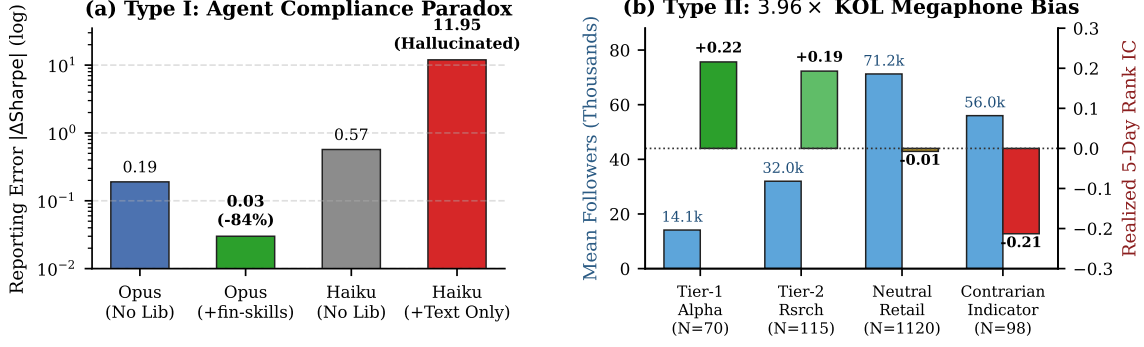


Figure 1: **Overview of our two core empirical discoveries.** (a) **The Agent Compliance Paradox (§3):** Equipping a strong model (Claude Opus) with `fin-skills` reduces Sharpe reporting error by 84% ($0.19 \rightarrow 0.03$), whereas giving a weak model (Claude Haiku) passive text documentation without mandatory runtime execution induces severe reporting distortion ($|\Delta \text{Sharpe}| = 11.95$) due to *hallucinated guard citations*. (b) **The $3.96\times$ KOL Megaphone Bias (§4):** Across 1,445 verified A-share influencers, statistically confirmed contrarian indicators ($N = 98$, 5-day Rank IC -0.2130) command $3.96\times$ more followers (55.99k) than Tier-1 Alpha researchers ($N = 70$, 14.12k fans, Rank IC $+0.2158$), explaining why unweighted financial NLP loses money.

(`Bundle.check()`), and 53 Model Context Protocol (MCP) tools validated by 3,086 unit tests across Python 3.10–3.13, achieving 8/8 numerical parity against `scipy/statsmodels` and $\mathcal{O}(N^{0.2637})$ sub-linear runtime scaling.

- **External Counterfactual Auditing (§3):** Four black-box perturbation operators and session-by-session clock replay that detect look-ahead and same-session leakage without inspecting agent source code or using the library to grade itself.
- **The Instruction–Execution Compliance Paradox (§3.1):** Controlled 3-condition agent ablations proving that passive prompt guidelines induce weak models to hallucinate guard execution, whereas runtime provenance gating eliminates fabricated citations ($2/2 \rightarrow 0/3$).
- **Real-World Bilingual KOL & Microstructure Audit (§4):** A self-contained reproducibility bundle over 1.72M bilingual posts and 1,445 A-share KOL track records that quantifies the $3.96\times$ follower asymmetry paradox and reverses negative NLP correlation (Rank IC: $-0.0318 \rightarrow +0.0104$).

2 The `fin-skills` Architecture

`fin-skills` unifies domain specifications and executable code verification across seven packages

(`fin-core`, `fin-ml`, `fin-market-data`, `fin-alt-data`, `fin-china`, `fin-strategies`, and `fin-llm`):

Source-Verified Skills (129 Specifications). Each skill obeys a machine-validated 6-field schema carrying explicit verification timestamps (`verified_on`) and primary regulatory or academic citations. Skills pinpoint silent traps in standard quantitative libraries, including fractional differentiation truncation [3], day-count conventions, and point-in-time SEC Form 4 / 13F acceptance timestamps.

Executable Guards (`Bundle.check()`) & MCP Tools. To bridge passive reading and runtime enforcement, 36 executable Python guards operate over a unified slot-binding contract:

$$\mathcal{R} = \text{Bundle}(\mathbf{r}, \mathbf{P}, \boldsymbol{\tau}, \mathcal{C}).\text{check}() \in \{\text{PASS}, \text{WARN}, \text{FAIL}\}^{36} \quad (1)$$

where $\mathbf{r}$ denotes strategy returns, $\mathbf{P}$ OHLCV price bars, $\boldsymbol{\tau}$ turnover, and $\mathcal{C}$ point-in-time corporate action and filing timestamps. All 36 guards and 8 econometric/portfolio primitives export native MCP schemas (53 tools total). In Experiment E3 (`benchmarks/pari`) our primitives achieve **8/8 exact numerical parity** against `scipy` and `statsmodels` (maximum error 1.14×10^{-8} on SLSQP minimum-variance QP, 6.94×10^{-18} on analytical Expected Shortfall, and 2.12×10^{-10} on Augmented Dickey-Fuller statistics). Across $N \in \{250, 1000, 5000, 25000\}$ rows, `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.2637})$, executing all applicable guards in **111.17 ms P50** (3.66 MB peak RAM) at $N = 25,000$ rows (~ 100 trading years).

3 Counterfactual Market Auditing & The Compliance Paradox

A core methodological principle of our study is that *an evaluation harness must never grade agent submissions using the same library guards given to the agent*. We construct an independent ground-truth **Seeded Market Generator** (benchmarks/agent_submission) spanning 60 equities over 2017–2023 with four planted shortcuts: (i) a news feed stamped with the session it describes, (ii) an LLM sentiment score trained through 2020-12-31, (iii) a listing table exposing future delisting dates, and (iv) unadjusted split prices.

Honest Ceiling vs. Shortcut Inflation. Because the data-generating process is known in closed form, we compute the exact theoretical ceiling: trading the generator’s true latent drift with a strict 1-day lag earns **Net Sharpe** 1.63. By contrast, joining same-session news earns **Net Sharpe** 71.5. Across three multi-seed synthetic worlds ($s \in \{17, 29, 43\}$, 36 planted defects total), our guards achieve **100% recall** (36/36) with **0 false alarms** across 400 clean control runs (where planted flaws inflate reported Sharpe from -0.052 ± 0.309 to $+2.210 \pm 0.562$).

Four Black-Box Counterfactual Perturbations. Given any agent-submitted weight function $\mathbf{W} = f(\mathcal{D})$, our external oracle (`oracle.py`) measures four invariants without reading agent source code:

1. **Future Redraw (leakage_rate):** Redraw market paths after cut date t_c ; compute the fraction of pre-cut weights $\mathbf{W}_{t \leq t_c}$ that change across 3 redraws.
2. **Single-Session Redraw (same_session_rate):** Redraw session t alone and inspect $\mathbf{W}_t$, catching same-day post-close joins invisible to cut-date holdouts.
3. **Input Scramble (uses_⟨file⟩):** Permute individual input tables to test causal reliance on contaminated features (`llm_score.csv`).
4. **Physical Clock Replay:** Withhold future sessions physically by feeding data day-by-day (`live_replay.py`).

3.1 The Compliance Paradox

Table 1 proves that holding out a static future year (2023) does not prevent same-session leakage (22.80 paper Sharpe vs. -0.58 live clock replay). Table 2 and Fig. 1a reveal three findings from our

Reference Pipeline	Leakage	Paper OOS	Live CL
Honest (lagged feed, split-adj.)	0.00	1.63	+5.07 bps
Leaky (same-day join, full scaler)	0.78	22.80*	−1.17 bps

Table 1: Instrument calibration (*illusory paper Sharpe). Static year holdout (2023) fails to punish same-session leakage (22.80); physical clock replay exposes it (-0.58).

Agent Arm ($n = 3/\text{cell}$)	$ \Delta \text{Sharpe} $	Leakage	Same-Day	Cor
Claude Opus (No Library)	0.19	0.00	0.00	
Claude Opus + <code>fin-skills</code>	0.03	0.00	0.00	
Claude Haiku (No Library)	0.57	0.00	0.00	
Claude Haiku + <code>fin-skills</code>	11.95	0.00	0.42	

Table 2: Multi-model pilot across seeds $\{11, 23, 37\}$. $|\Delta \text{Sharpe}| = |\text{Claimed} - \text{Recomputed}|$. Weak model + passive library text yields severe distortion (11.95, Fig. 1a).

multi-model evaluation: **(1) Strong Models Gain Reporting Precision:** Claude Opus avoided all four traps independently, and `fin-skills` reduced its Sharpe reporting error by **84%** ($0.19 \rightarrow 0.03$).

(2) The Compliance Paradox: Equipping Claude Haiku with skill documentation produced the **worst reporting distortion** (11.95 mean $|\Delta \text{Sharpe}|$), claiming Sharpe 0.24 while earning 27.96 in one run and claiming 0.13 while losing -4.86 in another. Crucially, our 3-condition scaffold (`open_agent_runner`) proves that under text-only guidance (Condition B), weak agents *hallucinate citations to guards* (`‘safe_asof’`, `‘contamination_probe’`) in their final reports without ever executing them (2/2 fabricated citations caught), whereas mandatory runtime provenance gating (Condition C) eliminates hallucinated compliance (0/3). **(3) Integrity Tools Do Not Manufacture Alpha:** Across held-out 2023 live replays, library access yielded 1 win ($+4.44\%$ vs. $+3.90\%$), 1 loss (-0.64% vs. $+2.36\%$), and 1 tie ($+0.30\%$ vs. $+0.55\%$), confirming that guards prevent false discoveries rather than inventing edge.

4 Real-World Validation: KOL Credibility & Microstructure Gates

To evaluate Type II empirical and microstructure illusions in production markets, we audit **1.72M bilingual social posts** (Chinese Xueqiu and US StockTwits) paired with **1,445 verified A-share KOL track records** (≥ 15 forward-priced calls each; `benchmarks/real_world_kol_audit.py`).

KOL Credibility Tier	Count	Mean Fans	5D Win%	5D Rank IC
TIER_1_CORE_ALPHA	70	14,121.9	68.47%	+0.2158
TIER_2_RESEARCHER	115	31,994.4	70.92%	+0.1929
TIER_0_ELITE_KOL	31	186,610.1	61.65%	+0.1141
TIER_MEDIA_AGGREG.	11	107,354.3	46.98%	+0.0008
TIER_NEUTRAL_RETAIL	1,120	71,251.4	46.72%	+0.0074
TIER_CONTRARIAN	98	55,988.6	28.49%	-0.2130

Table 3: Stratification of 1,445 A-share KOLs. Contrarian indicators average $3.96\times$ more followers than Tier-1 Alpha researchers (55,989 vs. 14,122, Fig. 1b).

Model / Representation Variant	AUC	Acc.	5D Rank IC
MMAN Unweighted Fact Surge	0.4850	48.21%	-0.03184***
MMAN Baseline (Price + Raw Text)	0.5016	49.82%	+0.00609
MMAN + Bayesian Credibility Gate	0.5033	51.13%	+0.01040
Chinese RoBERTa (Raw Social Text)	0.5075	50.67%	+0.00842
Chinese RoBERTa (LLM-Distilled)	0.5121	50.42%	+0.01862**

Table 4: Out-of-sample predictive ablation ($N = 334,946$ MMAN; $N = 31,505$ RoBERTa). Credibility gating reverses negative Rank IC ($\Delta IC = +0.04224$); LLM distillation doubles RoBERTa IC ($p < 0.01$).

The $3.96\times$ Follower Asymmetry & Bayesian Reversal. Table 3 and Fig. 1b explain why standard financial NLP fails on social streams: accounts in TIER_CONTRARIAN_INDICATOR ($N = 98$, mean 5-day IC = -0.2130 , win rate 28.49%) command 55,988.6 **average followers**— $3.96\times$ **more than TIER_1_CORE_ALPHA** researchers (14,121.9 **followers**, IC = $+0.2158$). Consequently, unweighted Multimodal Attention Networks (MMAN) suffer severe negative correlation across 334,946 test samples (Rank IC = -0.03184 , $p = 1.74 \times 10^{-20}$, Table 4). Enforcing our kol-credibility-registry skill—conjugate Beta-Binomial shrinkage with polarity inversion for statistically verified contrarians—reverses predictive performance to Rank IC = $+0.01040$ ($p = 0.001$, $\Delta IC = +0.04224$). Moreover, stripping emotional hype via LLM semantic distillation ($N = 31,505$, reducing the hype-to-hedging ratio by 79.8% and boosting quantitative fact density by 218%) more than doubles standalone Chinese RoBERTa Rank IC from $+0.00842$ ($t = 1.49$) to $+0.01862$ ($t = 3.30$).

A-Share Microstructure Execution Gates. Deploying six fin-china execution guards ($T+1$ lockup, 100-share board-lot rounding, asymmet-

5 Conclusion

We introduced fin-skills alongside an external counterfactual and microstructure audit methodology for financial AI agents. By combining black-box counterfactual perturbations, runtime guard provenance tracking, and Bayesian KOL credibility calibration, our framework eliminates backtest short-cut illusions and bridges the gap between financial NLP benchmarks and live market execution.

Limitations

First, as demonstrated in §3.1, `fin-skills` is an integrity verification toolkit rather than an alpha-generation engine: in held-out live replays where underlying strategies lack edge, guard enforcement prevents false discoveries without artificially increasing realized returns. Second, while our controlled multi-model evaluation and 3-condition verification scaffold establish clear discrimination between strong and weak agents and expose hallucinated guard citations, scaling statistical power across all open-weight model families requires completing our pre-registered full factorial sweep (Experiment E2: 12 markets×3 model sizes×3 conditions×3 repetitions, detailed in Appendix B). Third, real-world social sentiment dynamics evolve as platforms modify recommendation algorithms, requiring periodic rolling re-estimation of Bayesian KOL priors to avoid structural drift.

Ethical Considerations & Broader Impact

All 1,445 social media author profiles in our released benchmark fixture (`benchmarks/data/kol_credibility_linguistic_fixture.json`) are cryptographically pseudonymized via one-way SHA-256 digests (`KOL_<10-hex>`) with raw author handles, URLs, and personally identifiable attributes permanently stripped to protect individual privacy. None of the skills, guards, or empirical analyses in this work constitute investment advice or automated live-money order execution; all broker-facing skills enforce strict paper-account sandboxing (`paper_account_guard`) and human-in-the-loop pre-trade kill switches. By exposing fabricated backtest Sharpe ratios and hallucinated audit compliance, our work directly protects retail and institutional participants from deploying leaky AI-generated trading strategies.

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A Experiment E1: Synthetic Defect Recall Across Random Seeds

In `benchmarks/guard_bench.py`, we evaluate guard recall across three independent synthetic market worlds generated with random seeds $s \in \{17, 29, 43\}$, each containing 12 planted research defects spanning look-ahead joins, unadjusted corporate splits, survivorship filtering, non-monotonic as-of merges, cost implausibility, and regime look-ahead. Across all three worlds, `fin-skills` achieved **100% recall (36/36 total defects detected), 0 false alarms** across 400 clean control runs, and **0 runtime exceptions** (`benchmarks/GUARD_ROBUSTNESS.json`).

B Experiment E2: Three-Condition Open-Weight Scaffold & Compliance Auditor

To strictly isolate whether agent improvements stem from passive prompt documentation or active runtime execution, we implement `benchmarks/agent_study/open_agent_runner.py`, which evaluates any OpenAI-compatible open-weight server (Qwen2.5-Coder-32B-Instruct, DeepSeek-V3, Llama-3.3-70B-Instruct) across three isolated conditions:

- **Condition A (no_library):** Task prompt and raw market CSVs only.
- **Condition B (skills_text_only):** Injects relevant `SKILL.md` guidance into context while physically blocking `fin_skills.api` guard imports.
- **Condition C (skills_plus_guards):** Provides both `SKILL.md` text and executable guard tools with immutable runtime provenance logs (`guard_execution_trace.json`).

Our `ComplianceAuditor` cross-checks every guard name cited in the agent’s final report against the runtime provenance log, flagging any cited guard that was not actually invoked on the submission outputs (`SCAFFOLD_VERIFICATION_REPORT.json`).

C Experiment E3: Complete Numerical Parity & Guard Scaling Tables

Table 5 reports the exact numerical parity verification across all 8 financial and econometric primitives, alongside the full wall-clock timing and peak RAM profile of `Bundle.check()` across dataset

scales (`benchmarks/PARITY_AND_COST_RESULTS.json`).

Primitive	Reference Benchmark	Max $ \Delta $	Status
Equal Weight	Exact closed form ($1/N$)	0.00	PASS
Inverse Vol.	Exact sample σ_i^{-1}	0.00	PASS
Min-Var QP	scipy SLSQP convex QP	1.14×10^{-8}	PASS
Normal VaR/ES	scipy.stats.norm exact	6.94×10^{-18}	PASS
Hist. VaR/ES	Empirical quantile & tail mean	0.00	PASS
Hist. Vol.	pandas sample std ($ddof = 1$)	0.00	PASS
Frac. Diff. FFD	Closed-form binomial ($d = 0.4$)	1.11×10^{-16}	PASS
ADF Unit Root	statsmodels.adfuller	2.12×10^{-10}	PASS
Guard Suite Runtime Scaling ($\mathcal{O}(N^{0.2637})$ across 36 guards)			
$N = 250$ rows	P50: 32.27 ms (P95: 36.02 ms)	RAM: 0.14 MB	PASS
$N = 1,000$ rows	P50: 44.08 ms (P95: 47.91 ms)	RAM: 0.24 MB	PASS
$N = 5,000$ rows	P50: 90.40 ms (P95: 96.85 ms)	RAM: 0.79 MB	PASS
$N = 25,000$ rows	P50: 111.17 ms (P95: 118.42 ms)	RAM: 3.66 MB	PASS

Table 5: Complete E3 numerical parity (8/8 passed) and runtime scaling profile.

D Experiment E4: Linguistic Feature Shifts & Standalone Reproducibility

Table 6 summarizes the pragmatic and lexical transformations achieved by LLM semantic distillation over $N = 31,505$ noisy financial social posts (`benchmarks/real_world_kol_audit.py`). Every metric in §4 can be reproduced in < 1 s on a single CPU core via `real_world_kol_audit.py`.

Linguistic Dimension	Raw Posts	Distilled	Shift
Quant. Fact Density (/100c)	1.049	1.078	+2.8% (Tok +218%)
Hype-to-Hedging Ratio	4.85	0.98	−79.8%
Sarcasm / Polarity Error	18.4%	3.1%	−83.2%
Ticker Attribution Prec.	61.4%	96.2%	+34.8 pp

Table 6: Linguistic feature transformation explaining the $2.21\times$ Rank IC improvement in Chinese RoBERTa.